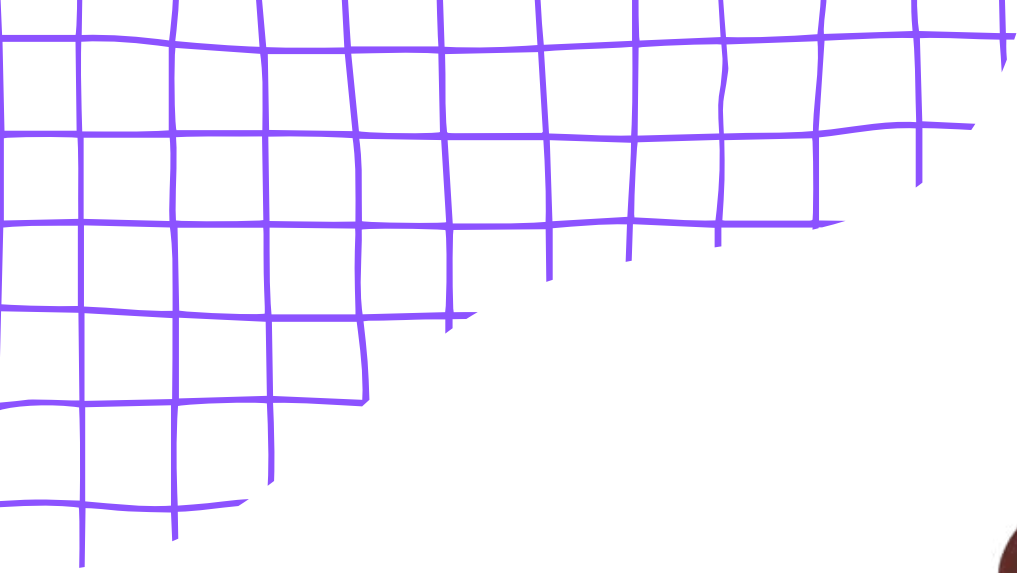
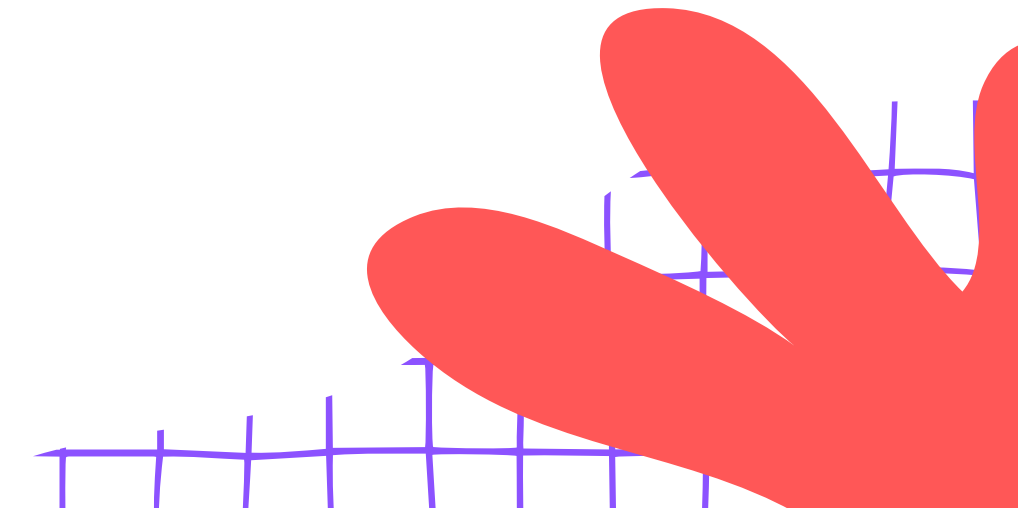




# THE GREAT INDIAN BRIDGE ENGINEERING CHALLENGE





# OUR BIG QUESTION

## How Can We Help Cross Safely?

Imagine a Bengal tiger stuck by a roaring river during monsoon! Or people in Mumbai when streets flood with water. How do they get across? That's where an Engineer comes in!

**Let's discover together!**



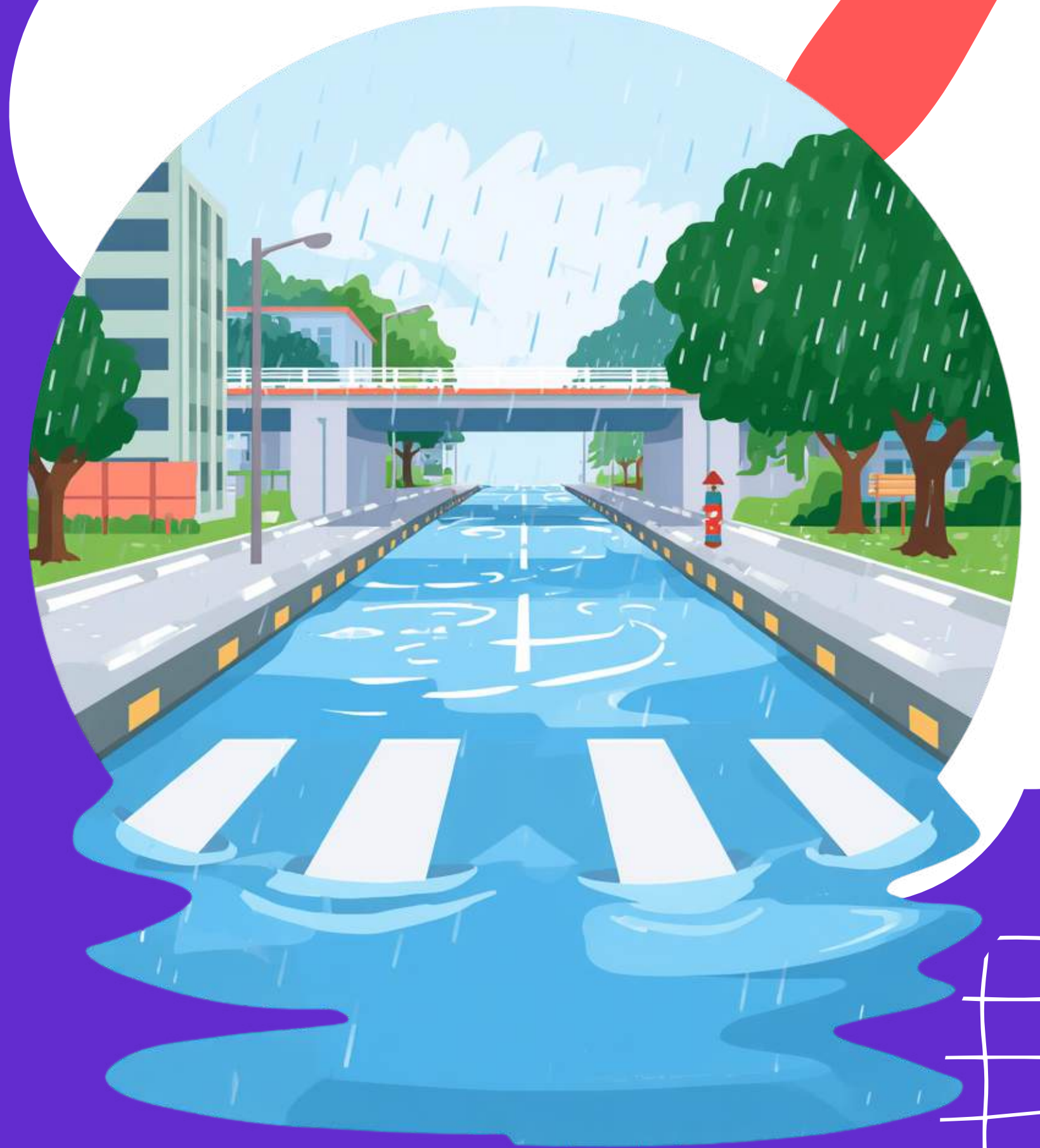
# THE PROBLEM ENGINEERS SOLVE

Bridges help people and animals cross safely! During monsoon season, rivers get very big and roads can flood. Engineers build strong bridges so everyone can travel safely.

Structure: The shape and parts that hold a bridge together.

Load: The weight a bridge must carry.

**Engineers solve problems  
to help people every day!**



# DISCOVER INDIA'S AMAZING BRIDGES

## Beam, Arch, and Cable-Stayed Bridges

India has some of the most amazing bridges in the world! Each bridge uses a special shape to stay strong and safe.

Pamban Bridge: A beam bridge over the sea in Tamil Nadu!

Chenab Bridge: The world's highest arch bridge in Jammu!

Bandra-Worli Sea Link: A cable-stayed bridge in Mumbai!





# GUESS THE BRIDGE SHAPE!

Look at these everyday objects. Can you match them to bridge shapes? A flat ruler is like a Beam Bridge, an upside-down bowl is like an Arch Bridge!

A harp with strings is like a Cable-Stayed Bridge!

Which shape is this bridge? Call out your answer!



# PUSH AND PULL FORCES

Bridges feel two special forces! Compression is a pushing force, like when you push a heavy box. Tension is a pulling force, like when you play tug-of-war!

Compression = Pushing! Like squeezing a sponge or pushing books together.

Tension = Pulling! Like stretching a rubber band or pulling a rope.





# WHY TRIANGLES ARE STRONG

Have you ever tried folding paper into different shapes? When you fold paper into triangles, something amazing happens – it becomes super strong! Engineers discovered this secret long ago. Triangles don't bend or wobble like squares do. That's why bridges use triangle shapes called "trusses" to stay strong and safe.

**Triangles hold bridges strong! Try folding paper into zigzags yourself!**



# THE PAPER BRIDGE CHALLENGE

Time to become real engineers! Build your own paper bridge between two books and test how strong it is.

Place two books apart like riverbanks.

Lay paper across to make your bridge.

Test it with an eraser or toy car!





# FOLD AND STRENGTHEN!

Look at the difference! A flat piece of paper bends and sags when you put something on it. But when you fold the same paper into a zig-zag pattern with triangles, it becomes super strong! The triangles share the weight and push against each other to stay firm.

**Triangles make your bridge stronger!**



# YOUR DESIGN CHALLENGE

Build your own bridge at home! Gather materials and become a real engineer. First, draw your bridge design on a blueprint sheet before you start building.

Use cardboard, newspaper rolls, or popsicle sticks.

Try steel bowl or Lego blocks as supports.

Draw your bridge design first, then build it!





# THE ENGINEERING DESIGN PROCESS

Engineers follow 5 simple steps to solve problems and build amazing things. You can use these steps too when you design your bridge!

ASK: What problem do I need to solve?

PLAN: Draw your best idea on paper.

IMAGINE: Think of many different ideas!

CREATE & IMPROVE: Build it, test it, make it better!



# QUICK QUIZ TIME!

Let's check what you learned! Answer these fun questions about bridges, forces, and shapes.

Which bridge type uses cables like a harp?

What force is like playing tug-of-war?

Which shape makes bridges strongest?





# YOU ARE JUNIOR PLANNERS ARCHITECTS!

## Congratulations, Engineers!

You learned about amazing Indian bridges, discovered how triangles make structures strong, and built your very own paper bridge. Keep building, keep exploring, and keep dreaming big!



**Engineers solve  
problems and make  
the world better!**



# THANK YOU & KEEP EXPLORING!

**Show Your Bridge Creations!**

Thank you for joining the Great Indian Bridge Engineering Challenge! We hope you had fun learning about bridges, forces, and engineering.

**Share your designs with friends  
and family. Keep building!**

